

Seminar 09: Auth & Observability in Practice

Let's wire up the app

The App

University course management — it works ✓

- View courses, students, enrollments
- REST API at `localhost:3000` · Frontend at `localhost:5173`

The problem:

- Anyone can create courses, enroll students — no auth
- Errors are silent — no monitoring

The Plan

- ① **Who are you?** → Clerk authentication
- ② **What can you do?** → RBAC authorization
- ③ **What broke?** → Sentry error tracking
- ◆ **Bonus:** OpenTelemetry traces + logs (*only after Phases 1–3*)

Prerequisites

Before writing any code:

1. **App running** — `bun dev` → <http://localhost:5173> shows courses ✓
 2. **Clerk account** — clerk.com · Email sign-in · add two keys to `.env`
 3. **Sentry account** — sentry.io · one **Node.js** project + one **React** project · add DSNs to `.env`
 4. **ClickStack open** — <http://localhost:8080> (*for OTEL bonus*)
- Full setup in the **Prerequisites** section of `ASSIGNMENT.md`

Phase 1 — Authentication

"Who are you?"

```
Browser
  ↓ sign-in form (Clerk hosted UI)
Clerk
  ↓ JWT in Authorization header
Elysia server
  ↓ verifyToken(token, { secretKey })
✓ continue / 401 Unauthorized
```

Tool: Clerk · **Steps** 1.1 → 1.4 in `ASSIGNMENT.md`

Phase 2 — Authorization

"What are you allowed to do?"

```
JWT payload
  ↓ contains permissions[]
requirePermission('org:courses:create')
  ↓ clerk.permissions.includes(permission)
✓ continue / 403 Forbidden
```

Configure roles & permissions in **Clerk dashboard** first.

Tool: Clerk RBAC · **Steps 2.1 → 2.4** in `ASSIGNMENT.md`

Phase 3 — Error Tracking

"What broke?"

```
Route handler throws
  ↓ onError catches it
Sentry.captureException(error)
  ↓ HTTPS → ingest.sentry.io
Sentry Issues dashboard
  ↓ grouped issue + stack trace + alert
```

Tool: Sentry · **Steps** 3.1 → 3.4 in `ASSIGNMENT.md`

Bonus — OpenTelemetry

"What is the system doing?"

```
Elysia request  
  ↓ @elysiajs/opentelemetry auto-instruments  
Spans (HTTP + DB + custom)  
  ↓ OTLPTraceExporter → localhost:4318  
ClickStack → traces view + logs view
```

Only attempt after Phases 1–3 are complete.

Tool: `@elysiajs/opentelemetry` · **Bonus steps 3.1 → 3.4** in `ASSIGNMENT.md`

Common Pitfalls

Clerk v6 — `<SignedIn>` / `<SignedOut>` are removed.

Use `<Show when="signed-in">` and `<Show when="signed-out">`.

Sentry + OTEL together — add `skipOpenTelemetrySetup: true` to `Sentry.init()`.

Without it, Sentry hijacks the TracerProvider and OTEL silently stops working.

ClickStack API key — no `Bearer` prefix.

Header is `Authorization: <key>`, not `Authorization: Bearer <key>`.

Let's Go

Open `ASSIGNMENT.md` · Start at **Prerequisites**

Ask questions early — don't wait until you're stuck